

$$\begin{array}{ccccc}
 Z_m(E) & \xrightarrow{\text{Laplace in } E} & Z_c(\beta) & \xrightarrow{\text{Laplace in } N} & Z_g(\mu) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Legendre in } \varepsilon = E/N & & \text{Legendre in } \rho = N/V & & \\
 \sigma = S_m/N & \longrightarrow & f = F/N & \longrightarrow & \Omega/V
 \end{array}$$